

Deep Networks

Convergence

Introduction to Deep Learning · Class 14

Outline

What Convergence Means

The Best Case: Equal Curvature

Ill-conditioning

Momentum

Learning-rate Schedules

Adaptive Learning Rates

Putting It Together

What Convergence Means

From a safe step to a fast one

Class 13 established that a step with $\eta < 2/L$ lowers the error. It said nothing about how long that takes.

Settled in Class 13

$-\nabla E$ is the steepest direction

$\eta < 2/L$ guarantees a decrease

the mini-batch gradient is unbiased

small batches settle in a noise ball

Open today

how many steps to reach the minimum

which η is best, not merely safe

what its noise costs in the limit

how to shrink that ball

Today: the best possible case first, where the answer is one step; then the property of real problems that destroys it, and four things we do about it.

Notation added today

$\lambda_{\min}, \lambda_{\max}$	extreme eigenvalues of $\mathbf{H}$	μ	momentum parameter
$\kappa = \lambda_{\max}/\lambda_{\min}$	condition number	β_1, β_2	Adam's decay parameters
α_i	coefficient along $\mathbf{u}_i$	$\mathbf{s}, \mathbf{r}$	running first and second moments
ρ	contraction factor per step	ϵ	stabilizing constant, $\sim 10^{-8}$

- Bishop writes the momentum parameter μ and Prince writes it β ; both appear in the literature

What it means for an iteration to converge

Definition (rate of convergence)

Write $\delta_T = E(\mathbf{w}^{(T)}) - E(\mathbf{w}^*)$. The iteration converges **linearly** (or geometrically) with factor $\rho \in (0, 1)$ if $\delta_T \leq \rho^T \delta_0$, and **sublinearly** if δ_T falls only as a power of T , for instance $\delta_T = O(1/T)$.

- “Linear” is about the *logarithm*: each step multiplies the error by a constant
- The number of steps to reach accuracy ϵ is then $\log(1/\epsilon)/\log(1/\rho)$
- Everything today is a fight to make ρ smaller

The tool: the quadratic model, again

Class 13 used the local quadratic model to say what a minimum looks like. The same model now says how the iterate moves on it.

$$E(\mathbf{w}) = E(\mathbf{w}^*) + \frac{1}{2} \sum_i \lambda_i \alpha_i^2, \quad \mathbf{w} - \mathbf{w}^* = \sum_i \alpha_i \mathbf{u}_i.$$

- $\mathbf{H}\mathbf{u}_i = \lambda_i \mathbf{u}_i$, and the $\mathbf{u}_i$ are orthonormal
- $\alpha_i = \mathbf{u}_i^\top (\mathbf{w} - \mathbf{w}^*)$ is the distance to the minimum along $\mathbf{u}_i$
- The analysis is exact for a quadratic and a good local description near any minimum

Bishop & Bishop (2024), Eqs. 7.7–7.11.

The Best Case: Equal Curvature

The directions never interact

Lemma 1 (Decoupled contraction)

Apply $\mathbf{w} \leftarrow \mathbf{w} - \eta \nabla E$ to the quadratic model. Then each coefficient evolves independently,

$$\alpha_i^{new} = (1 - \eta \lambda_i) \alpha_i^{old}, \quad \text{so} \quad \alpha_i^{(T)} = (1 - \eta \lambda_i)^T \alpha_i^{(0)}.$$

- Differentiating the model gives $\nabla E = \sum_i \lambda_i \alpha_i \mathbf{u}_i$.
- Write the update in the same basis, $\Delta \mathbf{w} = \sum_i \Delta \alpha_i \mathbf{u}_i$.
- Matching coefficients in $\Delta \mathbf{w} = -\eta \nabla E$ and using $\mathbf{u}_i^\top \mathbf{u}_j = \delta_{ij}$ gives $\Delta \alpha_i = -\eta \lambda_i \alpha_i$.
- Iterating that one-line recursion T times gives the stated power. □

Bishop & Bishop (2024), Eqs. 7.24–7.29.

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Bishop & Bishop (2024), Eqs. 7.24–7.29.

What that recursion already tells us

Corollary 2 (Stability and linear convergence)

The iteration reaches $\mathbf{w}^$ if and only if $|1 - \eta\lambda_i| < 1$ for every i , that is $0 < \eta < 2/\lambda_{\max}$, and it then converges *linearly* with $\rho = \max_i |1 - \eta\lambda_i|$.*

- Class 13 obtained $\eta < 2/L$ from smoothness; on a quadratic $L = \lambda_{\max}$, and the two agree exactly
- $1 - \eta\lambda_i$ may go *negative*: that is oscillation, harmless while the magnitude stays below one
- Convergence also needs every $\lambda_i > 0$, so the stationary point really is a minimum

The best case, stated

- If all λ_i are equal then $\eta = 1/\lambda$ makes every factor $1 - \eta\lambda_i$ exactly zero
- One step. $\rho = 0$. No method in the rest of this class improves on it
- The reason is that $\mathbf{H} = \lambda\mathbf{I}$, so $-\nabla E$ points **exactly** at the minimum from everywhere

Everything that follows exists because real error surfaces do not have equal curvature in every direction. The rest of this class measures that failure and repairs it.

Ill-conditioning

The condition number

Definition (condition number of the Hessian)

$$\kappa = \frac{\lambda_{\max}}{\lambda_{\min}} \geq 1.$$

A problem with large κ is called **ill-conditioned**.

- $\kappa = 1$ is the circular bowl of the previous section
- Geometrically, $\sqrt{\kappa}$ is the ratio of the longest to the shortest semi-axis of a contour ellipse
- $\lambda_{\max}$ limits how large η may be; $\lambda_{\min}$ limits how fast the slowest direction moves. One number cannot satisfy both

Bishop & Bishop (2024), §7.3.

The condition number as the shape of a contour

A contour of the quadratic model is the set $\frac{1}{2} \sum_i \lambda_i \alpha_i^2 = c$, where $\alpha_i = \mathbf{u}_i^\top (\mathbf{w} - \mathbf{w}^*)$. That is an **ellipse**: its axes point along the eigenvectors $\mathbf{u}_i$, and the semi-axis along $\mathbf{u}_i$ has length

$$a_i = \sqrt{2c} \lambda_i^{-1/2} \quad \Rightarrow \quad \frac{a_{\max}}{a_{\min}} = \sqrt{\frac{\lambda_{\max}}{\lambda_{\min}}} = \sqrt{\kappa}.$$

- The steeply curving direction gives the *short* axis: large λ , small $\lambda^{-1/2}$
- κ is therefore the elongation of the contour, squared

Bishop & Bishop (2024), Eqs. 7.10–7.14.

The slowest direction sets the pace

Proposition 3 (The rate at the stability limit)

Take η as large as stability allows, $\eta \rightarrow 2/\lambda_{\max}$. The fastest direction then barely contracts at all, and the slowest contracts by

$$\left|1 - \frac{2\lambda_{\min}}{\lambda_{\max}}\right| = \left|1 - \frac{2}{\kappa}\right| \quad \text{per step.}$$

For large κ this is $1 - 2/\kappa$: a factor that approaches one as the problem worsens.

- Pushing η to its limit buys speed in the direction that was never the problem; the long axis of the ellipse is the one that needs the steps

Bishop & Bishop (2024), Eq. 7.30.

The best a fixed step size can do

Proposition 4 (The optimal learning rate for a quadratic)

The choice $\eta^ = 2/(\lambda_{\min} + \lambda_{\max})$ minimises $\rho = \max_i |1 - \eta\lambda_i|$, giving $\rho^* = (\kappa - 1)/(\kappa + 1)$.*

- $\rho(\eta) = \max(|1 - \eta\lambda_{\min}|, |1 - \eta\lambda_{\max}|)$: **only the extremes can attain the maximum.**
- The first term falls in η , the second rises, so the maximum is smallest where they are equal.
- $1 - \eta\lambda_{\min} = \eta\lambda_{\max} - 1$ gives $\eta^* = 2/(\lambda_{\min} + \lambda_{\max})$.
- Substituting back gives the stated factor. □

Accuracy ϵ then needs about $\frac{\kappa+1}{2} \ln(1/\epsilon)$ steps: the cost is **proportional to κ** , however cleverly η is chosen.

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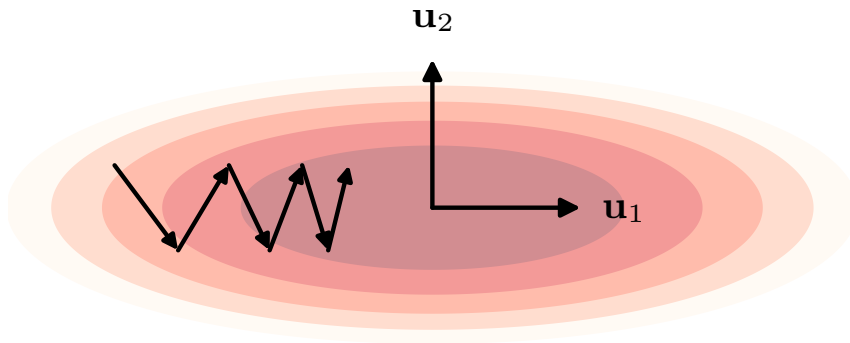
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The valley



An error surface whose curvature differs sharply with direction. At most points $-\nabla E$ is nearly perpendicular to the valley floor, so successive steps oscillate across it while progress along it is slow. u_1 and u_2 are the eigenvectors of the Hessian. Bishop & Bishop (2024), Fig. 7.3.

Where the condition number comes from

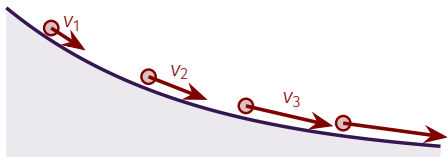
- **Input scale**: a feature measured in millimetres and one measured in kilometres produce curvatures differing by 10^{12}
- **Correlated inputs**: two nearly identical features make one eigenvalue nearly zero
- **Depth**: the same weight change has a different effect early and late in a network, so layers differ in curvature by orders of magnitude
- **Output scale**: an unnormalised target inflates every eigenvalue at once

Two responses are possible. Change the *iteration*, which is the rest of this class. Or change the *problem* so that κ is small to begin with, which is Class 15.

Momentum

A ball with inertia

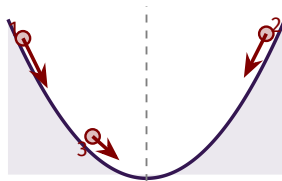
(a) a long, gentle slope



the slope pushes the same way each time,
so the pushes *add*

$v_1 < v_2 < v_3$: the ball builds speed

(b) a narrow gully



the slope reverses at every crossing,
so the pushes *cancel*

the sideways swing dies away

The same rule does both: it accelerates where the slope is consistent and stays out of the way where it alternates.

The analogy, term by term

A ball rolling downhill

it keeps some of its previous velocity

the slope adds to that velocity

a long consistent slope builds speed

a narrow gully cancels sideways motion

In the update

$$\mu \Delta \mathbf{w}^{(\tau-2)}$$

$$-\eta \nabla E$$

aligned gradients accumulate

alternating gradients cancel

- A ball does not stop and re-measure the slope at every instant, and neither should the iterate
- The single parameter μ decides how much of the past to keep

The momentum update

$$\Delta \mathbf{w}^{(\tau-1)} = -\eta \nabla E(\mathbf{w}^{(\tau-1)}) + \mu \Delta \mathbf{w}^{(\tau-2)}, \quad \mathbf{w}^{(\tau)} = \mathbf{w}^{(\tau-1)} + \Delta \mathbf{w}^{(\tau-1)}$$

- $\mu \in [0, 1)$; a typical value is 0.9
- Unrolling the recursion shows $\Delta \mathbf{w}$ is an exponentially weighted sum of **every** past gradient
- Prince writes the same idea as a running average, $\mathbf{m} \leftarrow \beta \mathbf{m} + (1 - \beta) \nabla E$ and $\mathbf{w} \leftarrow \mathbf{w} - \eta \mathbf{m}$, which rescales η by $(1 - \beta)$ but is otherwise identical

Bishop & Bishop (2024), Eq. 7.31; Prince (2023), Eq. 6.11.

Why momentum helps where it helps

Proposition 5 (Effective learning rate on a consistent slope)

If the gradient is essentially unchanged over a long run of steps, the momentum update converges to

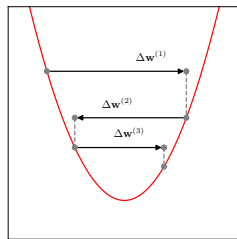
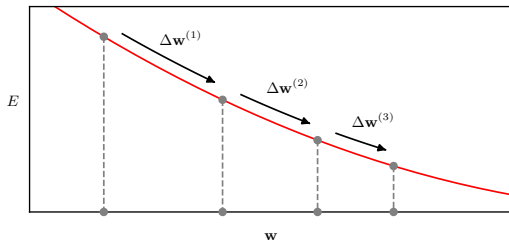
$$\Delta \mathbf{w} = -\eta \nabla E \{1 + \mu + \mu^2 + \dots\} = -\frac{\eta}{1 - \mu} \nabla E,$$

so the effective learning rate is $\eta/(1 - \mu)$. With $\mu = 0.9$ that is a tenfold increase.

- Where successive gradients **alternate** in sign the terms cancel instead, and the effective rate stays near η
- That asymmetry is the whole mechanism: acceleration along the valley, no amplification across it

Bishop & Bishop (2024), Eqs. 7.32–7.33.

The two regimes

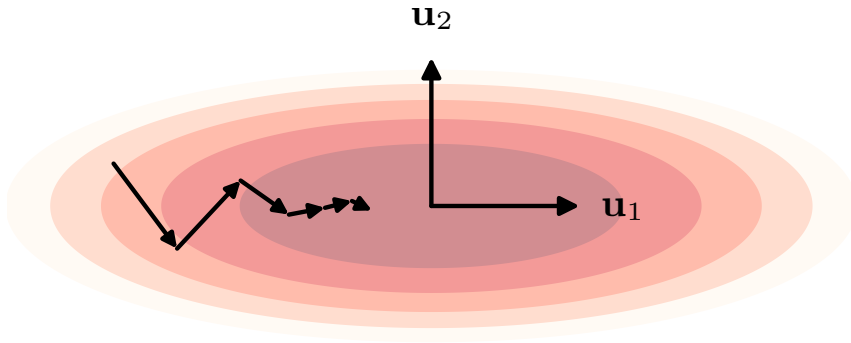


Left: on low curvature the updates point the same way and accumulate, raising the effective learning rate.

Right: where the steps oscillate the momentum contributions cancel and the effective rate stays near η .

Bishop & Bishop (2024), Figs. 7.4, 7.5.

Momentum on the valley



The same surface, now with a momentum term: the oscillation across the valley is damped and progress along it is much faster. Bishop & Bishop (2024), Fig. 7.6.

What momentum buys, precisely

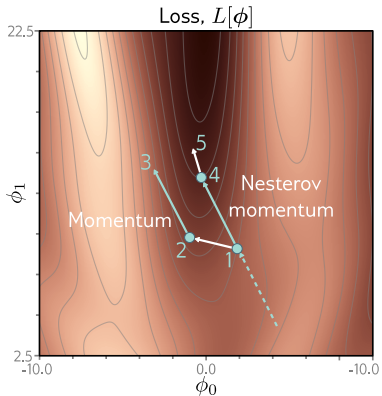
Proposition 6 (Heavy-ball acceleration on a quadratic)

On a quadratic with condition number κ , the choices $\mu^ = ((\sqrt{\kappa} - 1)/(\sqrt{\kappa} + 1))^2$ and $\eta^* = 4/(\sqrt{\lambda_{\max}} + \sqrt{\lambda_{\min}})^2$ give a contraction factor $\rho = (\sqrt{\kappa} - 1)/(\sqrt{\kappa} + 1)$.*

- Compare Proposition 4: κ has become $\sqrt{\kappa}$, so $\kappa = 10^4$ needs 100 times fewer steps
- The price is a second hyperparameter, and a trajectory that overshoots before it settles

Polyak (1964).

Nesterov momentum



$1 \rightarrow 2$ $-\eta \nabla E$ measured at 1
 $2 \rightarrow 3$ $+\mu \Delta \mathbf{w}^{(\tau-2)}$, the carry
 $1 \rightarrow 4$ $+\mu \Delta \mathbf{w}^{(\tau-2)}$ *first*
 $4 \rightarrow 5$ $-\eta \nabla E$ measured at 4

- Both routes use the same two vectors; only the **order** differs
- The look-ahead point 4 is past the turn, so the gradient there points back and corrects the overshoot within the same step: 5 lands nearer the minimum than 3

Prince (2023), Fig. 6.8.

The guarantee behind the name

Proposition 7 (Accelerated rates)

For an L -smooth convex E , Nesterov's method attains $E(\mathbf{w}^{(T)}) - E(\mathbf{w}^) = O(1/T^2)$, against $O(1/T)$ for plain gradient descent. If E is in addition strongly convex with condition number κ , the contraction factor is $1 - 1/\sqrt{\kappa}$ rather than $1 - 1/\kappa$.*

- No first-order method can do better than $O(1/T^2)$ using only gradients, so this is optimal in that class
- For **batch** gradient descent the improvement is real; with mini-batch noise it is much less clear-cut

Nesterov (1983, 2004); applied to SGD by Sutskever et al. (2013).

Stochastic gradient descent with momentum

Algorithm 8 (SGD with momentum)

Input: data indexed by n ; batch size B ; per-batch error $E_{n:n+B-1}$; learning rate η ; momentum μ ; initial $\mathbf{w}$

$n \leftarrow 1$; $\Delta \mathbf{w} \leftarrow 0$

repeat

$\Delta \mathbf{w} \leftarrow -\eta \nabla E_{n:n+B-1}(\mathbf{w}) + \mu \Delta \mathbf{w}$ *// update term*

$\mathbf{w} \leftarrow \mathbf{w} + \Delta \mathbf{w}$ *// weight vector update*

$n \leftarrow n + B$

if $n > N$ *then* shuffle data; $n \leftarrow 1$ *end if*

until convergence; *return* $\mathbf{w}$

Bishop & Bishop (2024), Algorithm 7.3.

Learning-rate Schedules

Why a fixed step size cannot finish the job

Momentum attacks the condition number. It does nothing about the noise in a mini-batch gradient.

- The mini-batch gradient's variance does not shrink as $\mathbf{w}$ approaches $\mathbf{w}^*$, so near the minimum the signal vanishes while the noise does not
- The iterate then wanders in a ball whose radius is set by η ; halving η halves the radius but never removes it
- Large η explores, small η refines — we want both, in that order

So make η a function of the step index: $\mathbf{w}^{(\tau)} = \mathbf{w}^{(\tau-1)} - \eta^{(\tau-1)} \nabla E_n(\mathbf{w}^{(\tau-1)})$.

Bishop & Bishop (2024), Eq. 7.35; Prince (2023), §6.2.2.

How fast may η be allowed to fall?

Proposition 9 (Robbins–Monro conditions)

For stochastic approximation with unbiased, bounded-variance gradients, the iterates converge to a stationary point provided $\sum_{\tau} \eta^{(\tau)} = \infty$ and $\sum_{\tau} (\eta^{(\tau)})^2 < \infty$.

- The first says the steps must still be able to travel an unbounded distance: decay too fast and the iterate stalls before it arrives
- The second says the accumulated noise must be finite: decay too slowly and it never settles
- $\eta^{(\tau)} \propto 1/\tau$ satisfies both; $1/\sqrt{\tau}$ fails the second

Robbins & Monro (1951).

Three algebraic schedules

linear

$$\eta^{(\tau)} = \left(1 - \frac{\tau}{K}\right)\eta^{(0)} + \frac{\tau}{K}\eta^{(K)}$$

power law

$$\eta^{(\tau)} = \eta^{(0)} \left(1 + \frac{\tau}{s}\right)^{-c}$$

exponential

$$\eta^{(\tau)} = \eta^{(0)} c^{\tau/s}$$

- The linear form decays over K steps and is then held at $\eta^{(K)}$
- $\eta^{(0)}$, $\eta^{(K)}$, K , s and c are found empirically
- Watching the learning curve is the practical test: it should be falling at a steady rate, not flattening early

Bishop & Bishop (2024), Eqs. 7.36–7.38.

Warm-up, and why the start is different

- Cosine annealing decays smoothly to zero over the whole run and has no schedule parameters to tune beyond its length
- **Warm-up** raises η from near zero over the first few thousand steps, then anneals
- The reason is that adaptive methods estimate gradient statistics from very few samples at the start, so those estimates are unreliable exactly when the steps are largest
- Warm-up is also standard when training with very large batches

Loshchilov & Hutter (2017); Goyal et al. (2018).

Adaptive Learning Rates

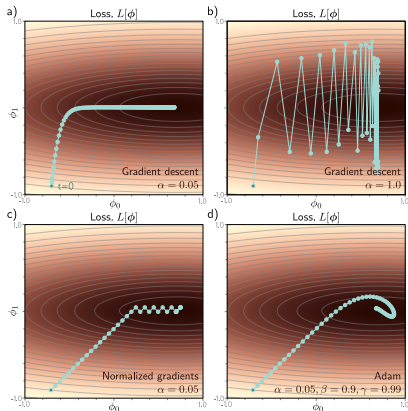
One rate cannot serve two curvatures

Momentum and schedules both keep a single η shared by every parameter. That is the next assumption to drop.

- Gradient descent makes **large** adjustments where the gradient is large — where it should perhaps be cautious
- And **small** adjustments where the gradient is small — where it should perhaps explore
- If the surface is far steeper in one direction than another, no single η makes good progress in both and stays stable
- The fix: give every parameter its own learning rate, adjusted automatically from what its gradients have been doing

Prince (2023), §6.4; Bishop & Bishop (2024), §7.3.3.

The problem, and three responses



- (a) an η suiting the steep direction leaves the flat one barely moving
- (b) an η suiting the flat direction is unstable
- (c) using only the sign moves a fixed distance along each axis, and cannot settle
- (d) Adam smooths both and does settle

Prince (2023), Fig. 6.9.

Normalising the gradient to its sign

$$\mathbf{m}^{(\tau+1)} = \nabla E, \quad \mathbf{v}^{(\tau+1)} = (\nabla E)^2, \quad \mathbf{w}^{(\tau+1)} = \mathbf{w}^{(\tau)} - \eta \frac{\mathbf{m}^{(\tau+1)}}{\sqrt{\mathbf{v}^{(\tau+1)}} + \varepsilon}$$

- The root and the division are **element-wise**, so every component reduces to its sign, and the step is a fixed distance η along each axis
- $\varepsilon \sim 10^{-8}$ prevents division by zero
- It progresses in every direction and **cannot converge**: it bounces around the minimum forever

Prince (2023), Eqs. 6.13–6.14.

AdaGrad

$$r_i^{(\tau)} = r_i^{(\tau-1)} + \left(\frac{\partial E}{\partial w_i} \right)^2, \quad \mathbf{w}_i^{(\tau)} = \mathbf{w}_i^{(\tau-1)} - \frac{\eta}{\sqrt{r_i} + \varepsilon} \left(\frac{\partial E}{\partial w_i} \right)$$

- Each parameter accumulates the sum of its own squared gradients, from $r_i^{(0)} = 0$
- Parameters that have seen large gradients — high curvature — have their rates reduced most
- **The flaw:** r_i accumulates from the first step and never forgets, so the steps eventually become too small and learning stalls

Duchi, Hazan & Singer (2011); Bishop & Bishop (2024), Eqs. 7.39–7.40.

RMSProp

$$r_i^{(\tau)} = \beta r_i^{(\tau-1)} + (1 - \beta) \left(\frac{\partial E}{\partial w_i} \right)^2, \quad \mathbf{w}_i^{(\tau)} = \mathbf{w}_i^{(\tau-1)} - \frac{\eta}{\sqrt{r_i} + \varepsilon} \left(\frac{\partial E}{\partial w_i} \right)$$

- One change from AdaGrad: the running *sum* becomes a running *average*, with $0 < \beta < 1$ and typically $\beta = 0.9$
- The estimate now forgets on a timescale of about $1/(1 - \beta)$ steps, so it tracks the curvature the iterate is currently in
- The learning rate can go back up when the surface flattens, which is exactly what AdaGrad could not do

Hinton (2012); Bishop & Bishop (2024), Eqs. 7.41–7.42.

Adam

$$\begin{aligned}s_i^{(\tau)} &= \beta_1 s_i^{(\tau-1)} + (1 - \beta_1) \frac{\partial E}{\partial w_i} & r_i^{(\tau)} &= \beta_2 r_i^{(\tau-1)} + (1 - \beta_2) \left(\frac{\partial E}{\partial w_i} \right)^2 \\ \hat{s}_i &= \frac{s_i^{(\tau)}}{1 - \beta_1^\tau} & \hat{r}_i &= \frac{r_i^{(\tau)}}{1 - \beta_2^\tau} & \mathbf{w}_i^{(\tau)} &= \mathbf{w}_i^{(\tau-1)} - \frac{\eta \hat{s}_i}{\sqrt{\hat{r}_i + \epsilon}}\end{aligned}$$

- RMSProp with momentum: an exponential average of the gradient *and* of its square. The name is short for **adaptive moments**
- Typical values are $\beta_1 = 0.9$ and $\beta_2 = 0.99$
- The most widely used training algorithm in deep learning

Kingma & Ba (2015); Bishop & Bishop (2024), Eqs. 7.43–7.47.

Why the bias correction is there

Proposition 10 (The averages start too small)

With $\mathbf{s}^{(0)} = 0$ and a gradient $\mathbf{g}$ that is constant in expectation,

$$\mathbb{E}[\mathbf{s}^{(\tau)}] = (1 - \beta_1^\tau) \mathbf{g},$$

so $\mathbf{s}^{(\tau)}$ underestimates $\mathbf{g}$ by exactly the factor $1 - \beta_1^\tau$. Dividing by that factor removes the bias, and since $\beta_1 < 1$ the correction tends to one as τ grows.

- **Unrolling gives** $\mathbf{s}^{(\tau)} = (1 - \beta_1) \sum_{k=0}^{\tau-1} \beta_1^k \mathbf{g}^{(\tau-k)}$.
- Taking expectations and summing the geometric series gives $(1 - \beta_1) \frac{1 - \beta_1^\tau}{1 - \beta_1} \mathbf{g}$.
- Dividing by $1 - \beta_1^\tau$ therefore returns $\mathbf{g}$ exactly; the same argument applies to r with β_2 . □

Bishop & Bishop (2024), Eqs. 7.45–7.46; Prince (2023), Eq. 6.16.

Why the bias correction is there

Proposition 10 (The averages start too small)

With $\mathbf{s}^{(0)} = 0$ and a gradient $\mathbf{g}$ that is constant in expectation,

$$\mathbb{E}[\mathbf{s}^{(\tau)}] = (1 - \beta_1^\tau) \mathbf{g},$$

so $\mathbf{s}^{(\tau)}$ underestimates $\mathbf{g}$ by exactly the factor $1 - \beta_1^\tau$. Dividing by that factor removes the bias, and since $\beta_1 < 1$ the correction tends to one as τ grows.

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Adam, as an algorithm

Algorithm 11 (Adam optimization)

Input: batch size B ; per-batch error $E_{n:n+B-1}$; learning rate η ; decay parameters β_1, β_2 ; stabilizer ϵ

$\mathbf{s} \leftarrow 0$; $\mathbf{r} \leftarrow 0$; $\tau \leftarrow 0$

repeat

choose a mini-batch at random; $\mathbf{g} \leftarrow \nabla E_{n:n+B-1}(\mathbf{w})$; $\tau \leftarrow \tau + 1$

$\mathbf{s} \leftarrow \beta_1 \mathbf{s} + (1 - \beta_1) \mathbf{g}$; $\mathbf{r} \leftarrow \beta_2 \mathbf{r} + (1 - \beta_2) \mathbf{g} \odot \mathbf{g}$ // element-wise

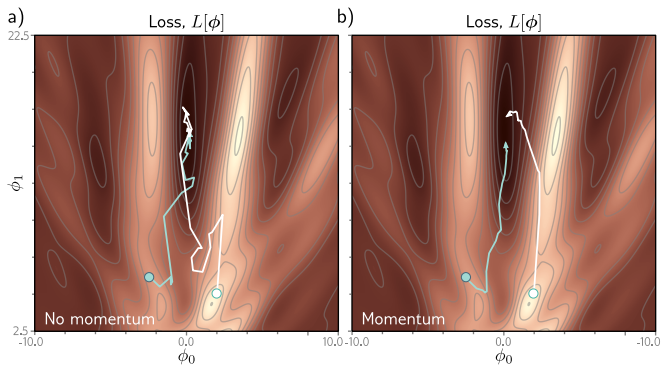
$\hat{\mathbf{s}} \leftarrow \mathbf{s} / (1 - \beta_1^\tau)$; $\hat{\mathbf{r}} \leftarrow \mathbf{r} / (1 - \beta_2^\tau)$ // bias correction

$\mathbf{w} \leftarrow \mathbf{w} - \eta \hat{\mathbf{s}} / (\sqrt{\hat{\mathbf{r}}} + \epsilon)$ // element-wise

until convergence; *return* $\mathbf{w}$

Bishop & Bishop (2024), Algorithm 7.4.

Momentum on a noisy problem



(a) stochastic gradient descent takes a very indirect path. (b) with momentum each change combines the previous one with the current batch gradient, which smooths the trajectory. Prince (2023), Fig. 6.7.

Four honest caveats

- The per-parameter argument assumes the principal curvature directions are **aligned with the axes** of weight space — a diagonal Hessian, which is unlikely. The methods work anyway
- Adam's original convergence proof was for convex losses; Reddi et al. (2018) exhibited a counterexample and proposed AMSGrad, which does converge
- Adam interacts badly with L^2 regularization; AdamW separates the two and is the common default
- Evidence runs in both directions on whether SGD with momentum generalizes better than Adam; much of the gap disappears once Adam's hyperparameters are tuned

Putting It Together

What each method actually fixes

Method	Attacks	Contraction factor
gradient descent	—	$(\kappa - 1)/(\kappa + 1)$
momentum	the condition number	$(\sqrt{\kappa} - 1)/(\sqrt{\kappa} + 1)$
Nesterov momentum	the condition number	$1 - 1/\sqrt{\kappa}$
schedules	the gradient noise	removes the floor
RMSProp, Adam	unequal curvature per parameter	no clean bound
normalization (Class 15)	κ itself	changes the problem

The first four change the *iteration* and leave the surface alone. Only the last changes the surface — which is why it is the subject of the next class.

The hyperparameters of the training algorithm

Choice	Typical	What it trades
optimizer	Adam, or SGD with momentum	speed against final quality
learning rate η	10^{-3} (Adam), 10^{-1} (SGD)	stability against progress
schedule	cosine, with warm-up	exploration against refinement
momentum μ	0.9	smoothing against responsiveness
β_1, β_2	0.9, 0.99	as above, per parameter
batch size B	a power of two	noise against cost (Class 13)

- These belong to the **algorithm**, not to the model, and are usually chosen by training several models and comparing on held-out data

Prince (2023), §6.5.

Takeaways

1. On a quadratic the eigen-directions contract independently by $(1 - \eta\lambda_i)$, so one η must serve every curvature at once (Lemma 1)
2. Stability is set by $\lambda_{\max}$ and speed by $\lambda_{\min}$; with equal curvature one step suffices
3. The condition number κ measures the gap, and the number of steps is proportional to it (Proposition 4)
4. Momentum accelerates along a consistent slope by $1/(1 - \mu)$ and cancels across an oscillating one (Proposition 5), replacing κ by $\sqrt{\kappa}$
5. A fixed η leaves a noise floor; a schedule satisfying $\sum \eta = \infty$, $\sum \eta^2 < \infty$ removes it (Proposition 9)
6. Adaptive methods give each parameter its own rate from its own gradient history; Adam is RMSProp with momentum and a bias correction

Next class: rather than repairing the iteration, we rescale the network so that κ is small to begin with.

References

- C. M. Bishop & H. Bishop, *Deep Learning: Foundations and Concepts*, Springer, 2024. §7.3 (Eqs. 7.24–7.47, Algorithms 7.3–7.4, Figs. 7.3–7.6).
- S. J. D. Prince, *Understanding Deep Learning*, MIT Press, 2023. §6.3–6.5 (Eqs. 6.11–6.18, Figs. 6.7–6.9).
- B. T. Polyak, “Some methods of speeding up the convergence of iteration methods”, *USSR Comp. Math.*, 1964.
- Y. Nesterov, “A method for solving the convex programming problem with convergence rate $O(1/k^2)$ ”, 1983.
- H. Robbins & S. Monro, “A stochastic approximation method”, *Ann. Math. Statist.*, 1951.
- J. Duchi, E. Hazan & Y. Singer, “Adaptive subgradient methods”, *JMLR*, 2011.
- D. P. Kingma & J. Ba, “Adam: a method for stochastic optimization”, *ICLR*, 2015.
- S. J. Reddi, S. Kale & S. Kumar, “On the convergence of Adam and beyond”, *ICLR*, 2018.
- I. Loshchilov & F. Hutter, “SGDR”, *ICLR*, 2017; “Decoupled weight decay regularization”, *ICLR*, 2019.

Questions?